



ESKISEHIR OSMANGAZI UNIVERSITY
ELECTRICAL AND ELECTRONICS ENGINEERING
2021-2022 SPRING SEMESTER
DIGITAL SYSTEMS LABORATORY

EXPERIMENT 1
TTL Gate Characteristics



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GROUP II – D

EXPERIMENT 1 REPORT

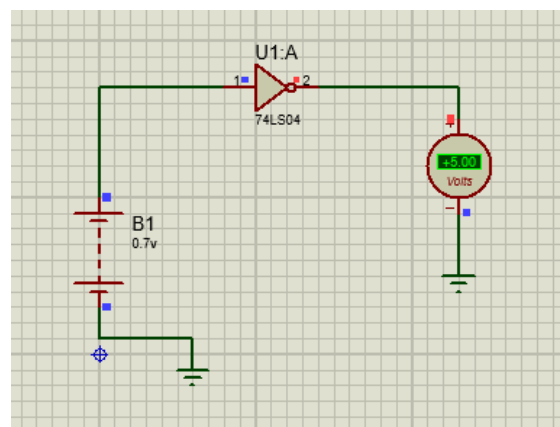
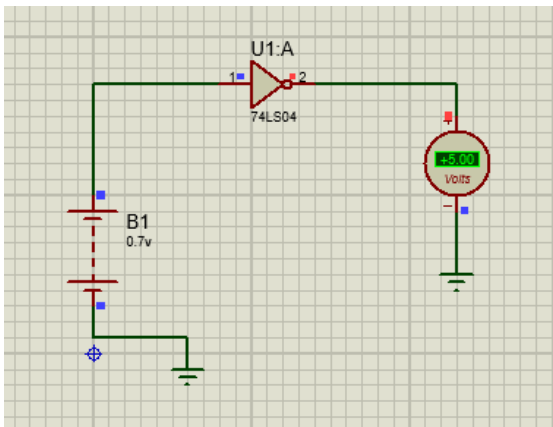
OBJECTIVES

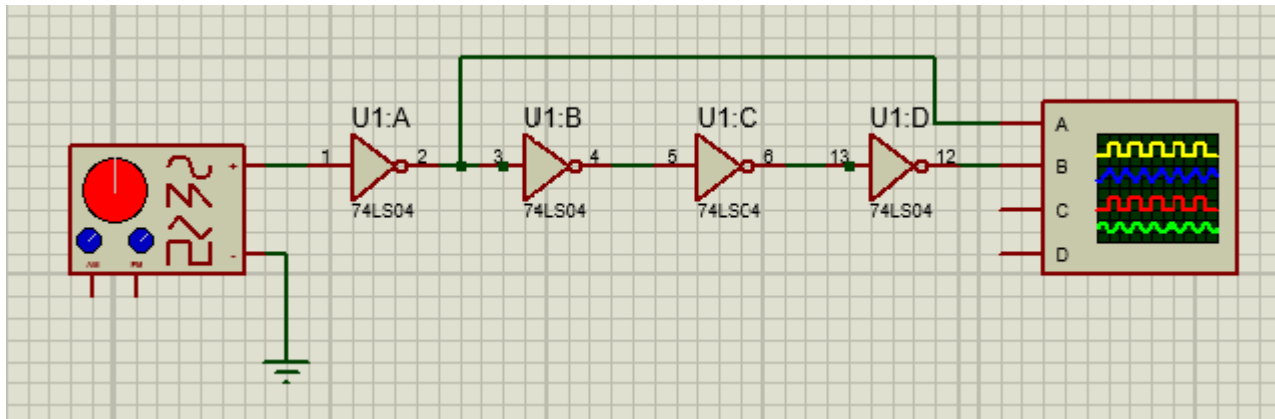
- Examination and use of the properties of TTL ports, input and output ports
- Voltage levels, noise margin etc. Examine different logic gate features including.

INTRODUCTION

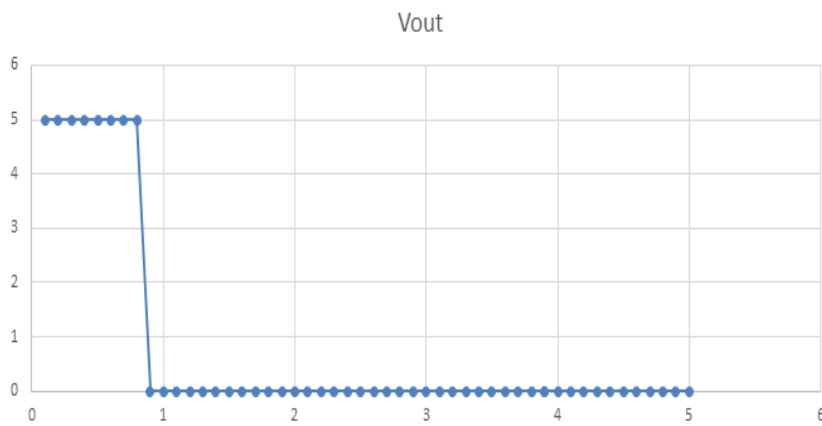
Three procedures were in the TTL Gate Characteristics experiment and all of them examined. Firstly is the voltage characteristics of the inverter. Second is calculating the noise margins for the gate using the voltage transfer curve. Third, the tPHL and tPLH of the in-line inverter were measured.

PROCEDURE





0.1 V	5 V	3.1 V	0 V
0.2 V	5 V	3.2 V	0 V
0.3 V	5 V	3.3 V	0 V
0.4 V	5 V	3.4 V	0 V
0.5 V	5 V	3.5 V	0 V
0.6 V	5 V	3.6 V	0 V
0.7 V	5 V	3.7 V	0 V
0.8 V	0 V	3.8 V	0 V
0.9 V	0 V	3.9 V	0 V
1 V	0 V	4 V	0 V
1.1 V	0 V	4.1 V	0 V
1.2 V	0 V	4.2 V	0 V
1.3 V	0 V	4.3 V	0 V
1.4 V	0 V	4.4 V	0 V
1.5 V	0 V	4.5 V	0 V
1.6 V	0 V	4.6 V	0 V
1.7 V	0 V	4.7 V	0 V
1.8 V	0 V	4.8 V	0 V
1.9 V	0 V	4.9 V	0 V
2 V	0 V	5 V	0 V
2.1 V	0 V		
2.2 V	0 V		
2.3 V	0 V		
2.4 V	0 V		
2.5 V	0 V		
2.6 V	0 V		
2.7 V	0 V		
2.8 V	0 V		
2.9 V	0 V		
3 V	0 V		



$$V_{IL}(\max) = 0.7V$$

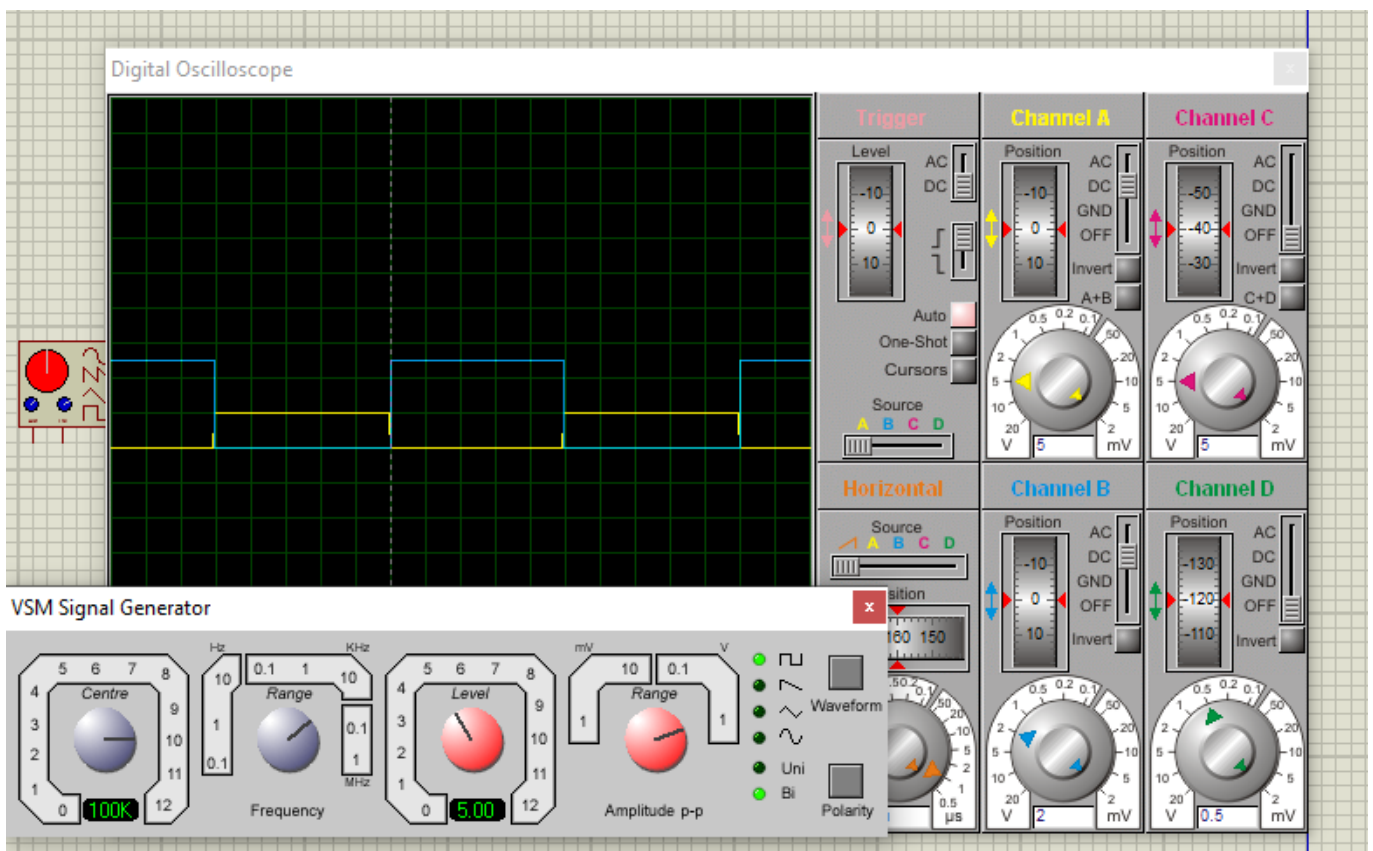
$$V_{IH}(\min) = 0.8V$$

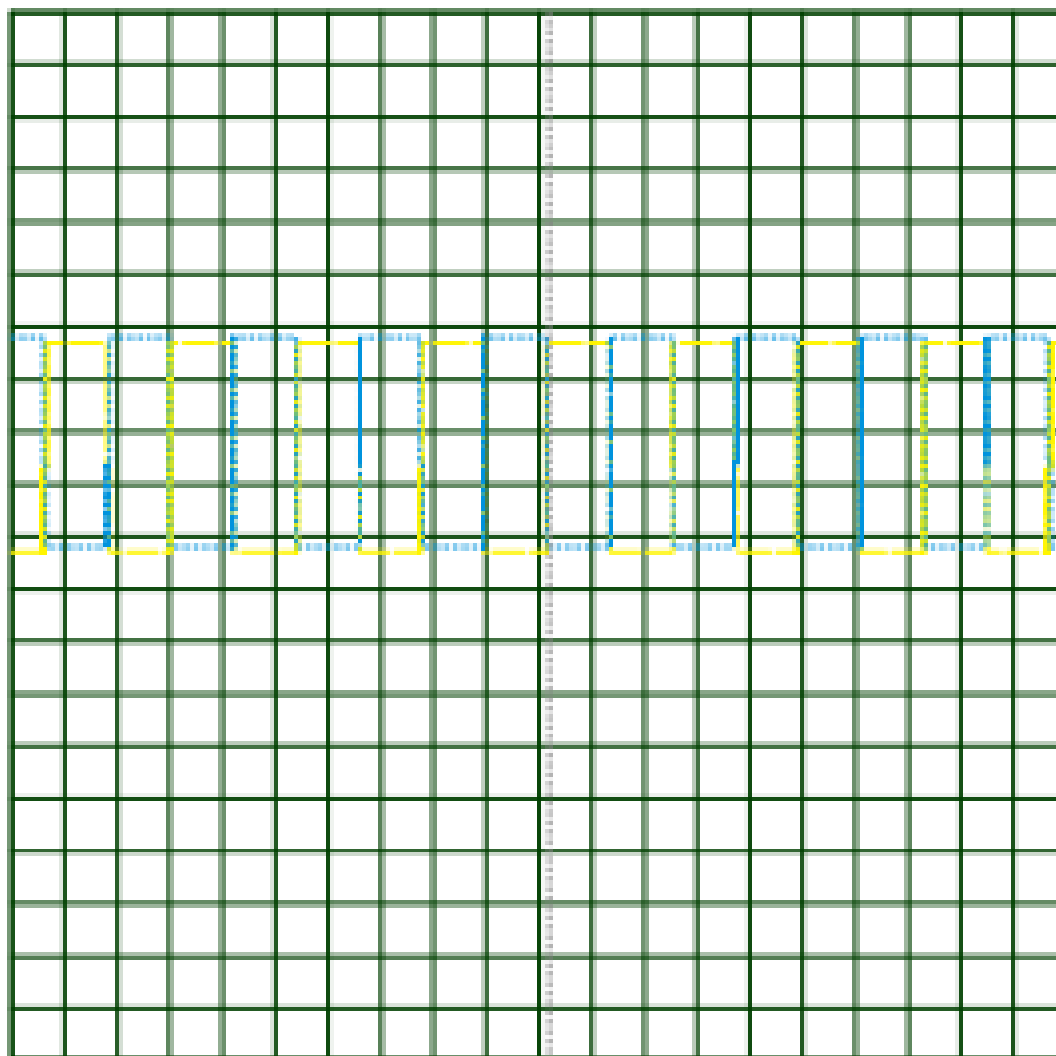
$$V_{OH}(\min) = 5V$$

$$V_{OL}(\max) = 0V$$

$$V_{NL} = V_{IL}(\max) - V_{OL}(\max) = 0.7V$$

$$V_{NH} = V_{OH}(\min) - V_{IH}(\min) = 4.2V$$





0

	Channel A	Channel B	Channel C	Channel D
V/Div	1.25 V	1.25 V	5.00 V	5.00 V
Offset	-750.00 mV	-500.00 mV	-20.00 V	-87.00 V
Invert	Normal	Normal	Normal	Normal
Coupling	DC	DC	Off	Off

	Horizontal	Trigger
Source	Trace	Channel A
Position	357.00 mS	Level 0.00 V
S/Div	35.00 mS	Coupling DC
		Edge Rising
		Mode Auto

DISCUSSION AND CONCLUSIONS

The output values of the 74ls04 inverter were examined from 0 to 5 Volts. The results were observed by increasing 0.1 from 0 to 5 volts. As we switched from 0.7 volts to 0.8 volts, the output voltage dropped from 5 volts to 0 volts. In addition, in the third procedure, the graph of the in-line inverter was examined together with the alternating voltage source on the oscilloscope and square waves were observed.